Subject Code: HTEH501

PAPER ID-311244

Roll No: | | | | | | | | | | | | | |

BTECH-HONOURS
(SEM V) THEORY EXAMINATION 2024-25
ENERGY STORAGE SYSTEM AND MANAGEMEN

TIME: 3 HRS **M.MARKS: 100**

Note: 1. Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. **2 x 10 = 20**

Qno.	Question	Marks	CO
a.	What is an energy storage system?	2	1
b.	What is meant by ultra capacitor?	2	1
c.	How many cells make battery?	2	2
d.	What are the parameters of electricity?	2	2
e.	What is battery pack system?	2	3
f.	What is a traction battery pack?	2	3
g.	What is the procedure of battery disposal and recycling?	2	4
h.	What are the chemical properties of a battery?	2	4
i.	What is the principle of solar photovoltaic?	2	5
j.	What is the principle of fuel cell?	2	5

SECTION B

2. Attempt any *three* of the following: **3x 10 = 30**

Qno.	Question	Marks	CO
a.	What is lead-acid battery acid? What is the working principle of lead-acid battery? Explain briefly.	10	1
b.	How do batteries convert chemical energy to electrical energy? What are ten uses of batteries? Explain briefly.	10	2
c.	How to choose a battery for EV? What are the criteria for battery selection in EV? Which battery used in HEV? Explain briefly.	10	3
d.	What is battery testing process? How do you test the battery? Explain briefly.	10	4
e.	What is the principle of wind energy? Explain its working with neat diagram.	10	5

SECTION C

3. Attempt any *one* part of the following: **1x 10 = 10**

Qno.	Question	Marks	CO
a.	What is Nickel based batteries? What is the working principle of Nickel based batteries? Explain briefly.	10	1
b.	What is Sodium based batteries? What is the working principle of Sodium based batteries? Explain briefly.	10	1

4. Attempt any *one* part of the following: **1x 10 = 10**

Qno.	Question	Marks	CO
a.	What are the variables to characterize battery operating conditions? Explain briefly.	10	2
b.	What is efficiency in a battery? Which battery has high efficiency? How to measure battery efficiency? Explain briefly.	10	2

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5. Attempt any *one* part of the following: 1x 10 = 10

Qno.	Question	Marks	CO
a.	What is the purpose of the battery monitoring system? Explain briefly.	10	3
b.	What does equalization do to a battery? How do you equalize battery cells? Explain briefly.	10	3

6. Attempt any *one* part of the following: 1x 10 = 10

Qno.	Question	Marks	CO
a.	What causes battery explosions? How do you prevent battery explosions? Explain briefly.	10	4
b.	What is the process of recycling EV batteries? Explain briefly.	10	4

7. Attempt any *one* part of the following: 1x 10 = 10

Qno.	Question	Marks	CO
a.	What is the function of a flywheel? Where are flywheels used? Why is a flywheel needed? Explain briefly.	10	5
b.	What is the hydrogen fuel cell? Why don't we use hydrogen fuel cells? Explain briefly.	10	5

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